

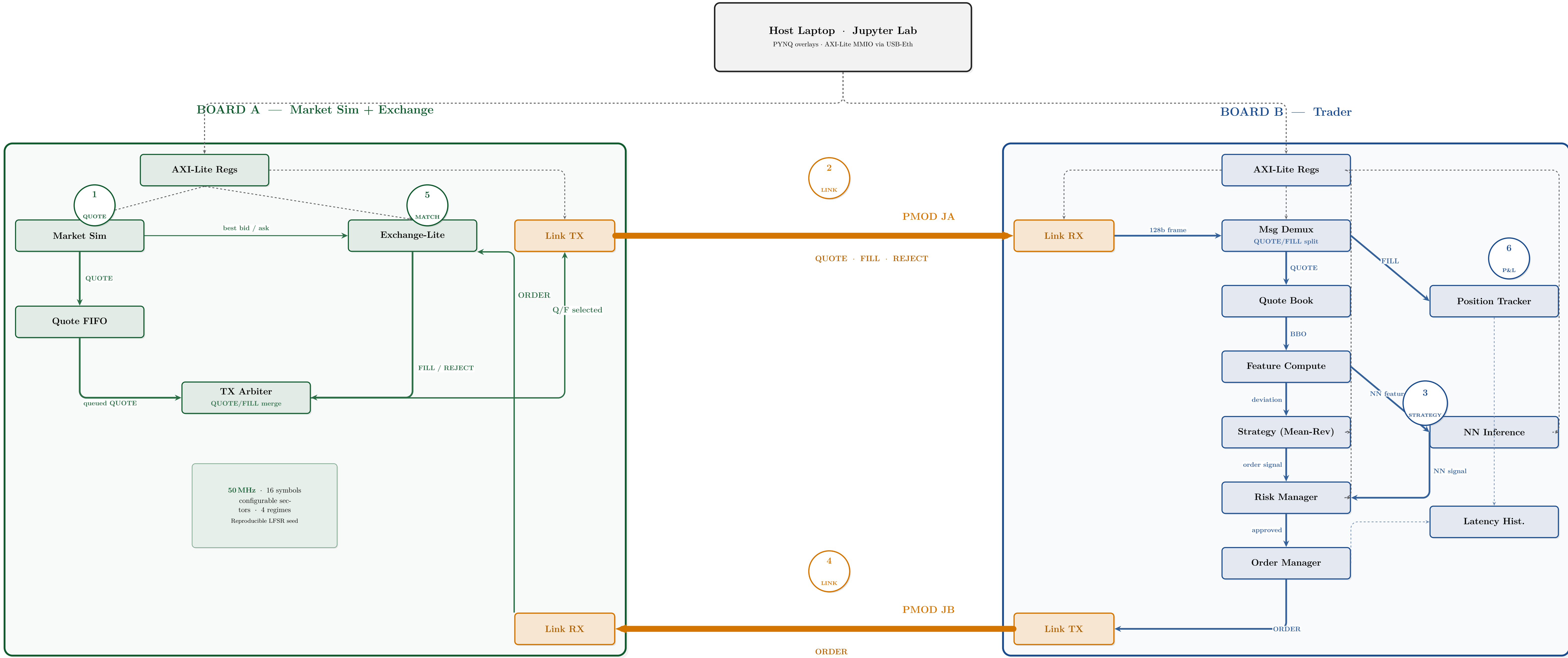
Introduction

Modern electronic trading needs **deterministic, sub-microsecond decisions**. A CPU running the same loop tail-jitters **10–100 μ s** from OS scheduling, cache misses, and interrupts — so the latency you measure in software is never the latency you actually got. Move the loop into **FPGA fabric** and every stage gets a fixed cycle cost, every metric is captured in hardware, and there is no OS in the path to lie about it.

TradeMark is two FPGAs on the bench: one is a synthetic stock exchange, the other is a trader, and they argue across a custom PMOD link at a **$\sim 2.07 \mu$ s round-trip (4-bit baseline)** with **deterministic cycle-level timing** — on *every* quote.

System Architecture

Round-trip: ① Quote → ② Link A→B → ③ Strategy → ④ Link B→A → ⑤ Match → ⑥ Fill & P&L



The Math Behind the Pipeline

Every quantity is **fixed-point**, every op is **shift-and-add**, every stage has a **deterministic cycle cost** — the discipline that makes hardware trading cycle-accurate.

1. Fixed-Point

Why every number is integer-encoded and why that gives us zero jitter.

Floats add unpredictable rounding. We use two scaled-integer formats so the FPGA does ordinary integer adds:

- Q16.16** (u32) for prices, spreads, EMAs — 16 integer bits (\$0–\$65,535), 16 fraction bits ($\sim 1.5 \times 10^{-5}$ resolution)
- Q32.16** (s48) for cash and P&L — handles $\pm \$2.1$ B before saturation
- 1-cycle** add/sub everywhere; no FPU $\Rightarrow$ no NaN/Inf, **zero jitter** between identical runs
- Same LFSR seed $\Rightarrow$ bit-exact replays for debug

2. Market Sim (A)

How 16 stock prices wander realistically without ever drifting away. Each tick, a price is pulled toward its anchor (mean-reversion) plus three layers of random shock:

$$p_{s,t+1} = p_{s,t} + \kappa_p (\bar{p}_s - p_{s,t}) + \sigma_r (n_g + n_{\text{sec}} + n_s)$$

pull to anchor 3-tier noise

κ_r, σ_r flip with the regime — **CALM** $\rightarrow$ **VOL** $\rightarrow$ **BURST** $\rightarrow$ **ADV**. Three LFSR-32s (period $2^{32}-1$) feed the noise. Bid/ask = $p_s \pm \frac{1}{2} \text{spread}_s$.

3. Strategy + Risk + NN (B)

Mean-reversion path plus quantized NN inference in parallel.

- Mean-Rev core**
 $\mu_{t+1} = \mu_t + \alpha(p_t - \mu_t)$
- Decision rule**
 $\Delta = p - \mu$; BUY if cheap, SELL if rich, HOLD near fair value
- Risk gates**
 $|q + \delta| \leq Q_{\max}, r_t \leq r_{\max}, C_t \geq -L_{\max}$
- NN path**
 $x \in R^9 \rightarrow 9 \rightarrow 128 \rightarrow 128 \rightarrow 64 \rightarrow 3 \rightarrow \{\text{HOLD, BUY, SELL}\}$
- Fixed-point NN**
Int8 W, Int16 activations, Int24 MAC; **4-cycle, 1/cycle**
- P&L update**
 $\text{P\&L}_t = C_t + \sum_s q_s p_s$

4. Latency

Why we can quote a single number and trust it.

Every stage has a fixed cycle count, so the round-trip is just their sum (no OS, no contention):

$$T_{\text{RT}} = \underbrace{1}_{\text{quote}} + \underbrace{32}_{\text{A} \rightarrow \text{B pipe}} + \underbrace{6}_{\text{B} \rightarrow \text{A pipe}} + \underbrace{32}_{\text{B} \rightarrow \text{A match}} + \underbrace{1}_{\text{P\&L}} \text{ cy}$$

@ 50 MHz (20 ns/cy): $T_{\text{RT}} \approx 2.07 \mu$ s (4-bit baseline), **every quote, no exceptions**. The on-chip Latency Histogram (16 $\times$ 8 cy bins) measures $\Delta t = t_{\text{fill}} - t_{\text{send}}$ in fabric, so the metric itself is jitter-free.

Hardware Implementation

Two AMD Zynq UltraScale+ **XCZU3EG** boards, all logic written in SystemVerilog around a shared **hft_pkg.sv**, every register-mapped surface driven from Python over AXI-Lite MMIO.

Board A — Market Sim + Exchange

Generates 16 correlated stock prices and matches incoming orders against best bid/ask in 1 cycle.

- Market Sim:** 3-tier LFSR noise (global / sector / per-symbol) with O–U pull-back; 4 regimes (CALM $\rightarrow$ VOL $\rightarrow$ BURST $\rightarrow$ ADV)
- Quote FIFO + TX Arbiter:** buffers quotes, prioritizes fill/reject responses for fair link bandwidth
- Exchange-Lite:** 1-cycle order match against live best bid/ask; emits FILL or REJECT
- AXI-Lite regs:** per-symbol mid, spread, sector, 16-bit company tokens — all writable live from Python

FSM: RESET $\rightarrow$ IDLE $\rightarrow$ RUNNING $\leftrightarrow$ STOPPED

Link Layer — Custom PMOD

Full-duplex 128-bit framed bus over two PMOD ribbons — our own protocol, no IP-core black boxes.

- Physical:** 2 $\times$ 12-pin PMOD (JA: A $\rightarrow$ B, JB: B $\rightarrow$ A) — simultaneous quote and order paths
- Frame:** 128 b = 4-b msg_type + 4-b sym_id + 32-b bid + 32-b ask + 16-b qty + 16-b seq + 24-b rsvd/CRC
- Mesochronous CDC:** each board has its own crystal; 2-FF synchronizer at every RX nibble deserializer
- Throughput:** 32 nibbles $\times$ 20 ns = **640 ns/frame**; 1.5 M frames/s ceiling, 64-deep TX FIFOs absorb burst storms

Handshake: ready/valid on both ends, no dropped frames

Board B — Trader Pipeline

7-stage deterministic pipeline; one quote in $\rightarrow$ one order out, in a fixed cycle count every time.

- Demux + Quote Book:** routes by msg_type, maintains live best bid/ask for all 16 symbols
- Feature Compute:** single-DSP per-symbol EMA fair value, deviation $\Delta = p_s - \mu_s$
- Strategy + Risk:** Mean-Rev (default) + NN inference mode; 3 parallel comparators for position, rate, drawdown caps
- Position Tracker + Latency Hist:** Q32.16 cash + per-symbol qty; on-chip 16-bin quote-to-fill measurement

FSM: B_RESET $\rightarrow$ B_IDLE $\rightarrow$ B_ARMED $\rightarrow$ B_TRADING $\leftrightarrow$ B_HALTED

Conclusions & Key Performance

$\sim 2.07 \mu$ s
Round-Trip Baseline
4-bit PMOD, wire-dominated

- Closed-loop dual-FPGA system:** Board A synthetic exchange + Board B trader over 128-bit PMOD framing
- Quote speed:** up to ~ 1.47 M frames/s per direction (4-bit PMOD, 680 ns min frame period, ready/valid backpressure)
- Market workload:** 16 symbols, 4 live-switchable regimes, sector-aware 3-tier LFSR + O–U style price dynamics
- Deterministic telemetry in fabric:** latency histogram + min/max/sum/count + per-symbol P&L/risk state
- Reproducible debug runs:** fixed-point datapath + seedable LFSR market stream enable bit-exact replay
- NN strategy integrated in RTL:** 9-feature quantized inference path (4-cycle latency, 1 decision/cycle)
- Link headroom at target load:** 100K qps $\approx$ 6.8% utilization; 1M qps $\approx$ 68% utilization (4-bit path)
- End-to-end closed loop in hardware:** QUOTE $\rightarrow$ ORDER $\rightarrow$ FILL path runs continuously across both boards in real time

2+4

Strategies + Market Modes
Mean-Rev + NN across CALM/VOL/BURST/ADV

~ 1.47 M/s
Max Quotes Processed
per direction (4-bit PMOD baseline)

- Project scale:** 27 SystemVerilog RTL modules + 26 SystemVerilog testbenches + 14 Python control/monitor tools
- Verification stack:** Python golden model $\rightarrow$ SV TBs $\rightarrow$ PYNQ on-board checks
- Clocking model:** 50 MHz trading clock in this build, mesochronous CDC at RX
- Deployment:** full PYNQ overlays + runtime control/monitor pipeline during live trading
- Run-time knobs:** strategy select, threshold, EMA alpha, position/rate/loss limits, symbol config
- Link robustness:** ready/valid backpressure + deep TX FIFOs prevent drops during bursty regimes
- Strategy architecture depth:** mean-reversion and quantized NN paths run in hardware with shared risk gating + order flow
- Observability depth:** per-symbol quote/position/P&L/EMA/signal snapshots exposed over AXI for live dashboarding

On-Fabric

Telemetry
latency hist + P&L + Risk counters